

### Lab 3: Log File Analysis

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## Executive Summary

This lab was focused primarily on log file analysis which can be super useful when it comes to digital forensics and administrators of systems. We looked at logs within Windows and Linux and reviewed their functions and purposes within the two operating systems, as well as their similarities and differences.

In the next part of the lab, we reviewed different log analysis tools including Windows Event Viewer, KSystemLog, and Splunk. Each of these tools have their own benefits and drawbacks when it comes to analysis.

In the final part of the lab, we analyzed log files on both Windows and Linux using the tools reviewed in the previous section. Scripts were run to make changes to the log files and then we had to find the different changes caused by them and identify why they could be a sign of intrusion.

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### Lab 3: Log File Analysis

For cybersecurity, logs can be used to determine what has occurred on a system. “Logs contain vital information such as the exact time and date an event happened, the device or application where the event originated, the user associated with the activity, what the user or process did (logging in, accessing a file, changing settings, etc.), and whether the action was successful or failed” (Dunne). The organization of these logs vary from Windows to Linux. Even so, no matter which operating system you use, there are options for log viewing and analysis. This lab focuses on log file analysis and the tools used to accomplish that task. “Log File Analysis is a powerful process used to examine log files generated by systems, networks, or applications. The primary purpose is to identify, track, and understand patterns, anomalies, and the overall system health, aiding businesses in making more data-driven decisions” (Dremio).

Starting off with Windows log files, they are commonly stored in the C:\WINDOWS\system32\winevt\Logs\ folder as .evtx files. Windows primarily uses logs for troubleshooting when problems arise. “These logs are essential for troubleshooting, monitoring system performance, and auditing purposes. They help system administrators and users track various activities, including system errors, security events, application errors, and system performance issues” (AlexHost). Windows logs

are stored in binary so without the help of an application to view them, they aren't easily examined, but luckily Windows has a built-in log viewer. Windows is a centralized logging system, which means that the logs are easily found in the same place (Event Viewer). Windows classifies logs as system logs, application logs, security logs, setup logs, and forwarded event logs. System logs keep track of system operations and are used to monitor the system and its performance. Application logs are provided by software during use and can be used by administrators to monitor a user's activity. Security logs focus on the security of the system and can be very valuable when investigating incidents since they keep track of access, policy changes, and other related events. Setup logs come from the installation of software and operating systems and can help resolve installation issues. Forwarded event logs come from other systems, specifically remote systems. These logs are all shown in Figure 1. Windows provides event IDs to help users identify different types of events that could be critical to cyber security, Figure 2. For example, Event ID 1102 is considered a high criticality because the audit log was cleared. This could indicate someone has done something on the system that they don't want the administrator to see. Microsoft has an entire page dedicated to its IDs and the potential criticality of it so it's easily accessible to get more information, though Event Viewer organizes all of it and provides a description so it isn't necessary for the average person.

Traditionally, Linux used rsyslog and the /var/log directory. However, Linux made the shift to system-journald and logs are now stored in a different manner with syslog being an optional addon for logging features. With this change, Linux shifted away from a decentralized logging system to a centralized system like Windows has. While

decentralized has its benefits for debugging purposes, for the average user, it provides more of a disadvantage because logs are dispersed in subdirectories. The journal is kept in binary and can't be viewed with normal terminal commands like `cat` or `grep`. In order to read logs, the command `journalctl` is used to display the logs in the terminal. Logs are assigned a priority level, much like Windows (discussed later on), although `systemd` provides more levels: emergency, alert, critical, error, warning, notice, informational, debug. "Systemd-journald is a system service that collects and stores logging data. It creates and maintains structured, indexed journals based on logging information that is received from the kernel, from user processes via the libc syslog call, from standard input and standard error of system services or via its native API" (Linux Commands). An example of the journal is shown in Figure 3. In Linux, `sudo` is how users can execute commands with superuser privileges. If there is an unexpected use of this, it could be an indication that there is a problem.

Despite Windows and Linux being two very different operating systems, there are some features with logs that appear in both. Login and logout events are logged when any attempt to is made and whether it was a success or failure. Repeated failed access attempts can indicate a brute-force attempt on a system. Reboots and shutdowns are logged for monitoring purposes such as security auditing and performance analysis. User privilege escalation is logged to track misuse of admin privileges. Filtering is possible on both operating systems which makes incident response much faster.

Logs can be viewed in Event Viewer on Windows machines. "Windows Event Viewer displays the Windows event logs. Use this application to view and navigate the logs, search and filter particular types of logs, export logs for analysis, and more"

(Loggly). This is a helpful tool because you are able to do everything you need to do in one application. There are five types of events on a Windows computer and they are: error, warning, information, success audit, and failure audit. These help an administrator determine how serious a problem is, as well as, where the problem is coming from. For example, if the device is low on space, it will likely be logged as a warning to let the administrator know that a problem will likely occur with space in the future. This is so the administrator can prepare in whatever way necessary, like uninstalling programs or adding more space, effectively solving the problem before it ever occurs. "Event Viewer allows you to filter events using a variety of parameters, including date, event level, source, and more" (Gouda). Filtering can help administrators narrow down logs to look at, whether it be during a security event or for a normal audit. It can decrease time loss and resources used.

KSystemLog is a Linux exclusive log viewer tool. "KSystemLog helps users understand what their machine does in the background. KSystemLog aims to simplify the reading of system log files. This program is developed for beginner users who don't know how to find information about their system or where the log files are located in their computer" (kde.org). KSystemLog offers real time monitoring and can organize logs so it's easy to filter and review. Instead of using commands in the terminal, KSystemLog makes it so there is a GUI which makes it easier for the user through visualization. KSystemLog categorizes logs by types, which are: authentication, daemon, kernel, system, x.org, cron jobs, boot logs, and apache. Categorization makes it easier to look at only what you need and declutter. Logs are also highlighted based on the level it's assigned. Real time monitoring is more beneficial for pen testing but can

also help with software troubleshooting. KSystemLog offers keyword filtering so certain strings can be searching, such as sudo or failed. This software is also able to email selected logs to the user for further investigation if needed. KSystemLog is shown in Figure 4. This tool was super organized and made it a little easier to read while keeping it simplistic.

Splunk is a popular SIEM, or security information and event management. Splunk “produces software for searching, monitoring, and analyzing machine-generated data via a web-style interface. Its software helps capture, index and correlate real-time data in a searchable repository, from which it can generate graphs, reports, alerts, dashboards and visualizations” (Wikipedia Contributors). The benefits of a SIEM are that it’s centralized, has threat detection capability, real-time alerts, and long-term log retention. Splunk is on the lighter end of SIEMs and is popular in the business world. It costs a fair bit of money to get the full extent of its usefulness and the more data, the more expensive Splunk can get so it may not be the most optimal choice on a budget. Splunk is capable of detecting brute force attacks, privilege escalation, lateral movement, and data exfiltration. Not only that but Splunk utilizes role-based access control for it’s dashboard and data so view of sensitive data can be limited.

Between the three, they could all be useful depending on the situation. My last choice would be Windows Event Viewer because I personally don’t like it. In a situation where there is a high volume of data, I believe that Splunk would be more beneficial because it has a lot more customization and it’s available across multiple platforms and not just one operating system. Splunk is also more geared toward security, though both Windows Event Viewer and KSystemLog are for local security. The perks of Windows



Event Viewer and KSystemLog are that they are both free and built-in (with just and install needed for KSystemLog). This makes them easily accessible to the average person and not for just enterprises. All three are capable of capturing logs in real time though Windows Event Viewer requires a refresh. Splunk has the capability to index logs in real time and alert as well, which is again much more useful for a business than a singular person. Each tool has its benefits when it comes to cybersecurity and they could all get the job of log file analysis done.

For the log analysis portion of this lab, I was a little bit confused to be honest. I started with Kali Linux and ran 'sudo ./lab3s3.sh'. The terminal window did not close but the file on the desktop disappeared so I assume that it ran. The script made a few things happen, which are shown in Figure 5. It created four new groups and four new users, one for each group. The new users (and groups) are Tyler, Dade, Elliot, and Thomas. Each one's password was changed. Elliot and Thomas both had their privileges escalated and were added to group sudo and shadow group sudo. Next, I ran 'sudo ./lab3s4.sh', results shown in Figure 6. This script deleted all four of the users and removed the groups made for each of them. Thomas and Elliot were both removed from the sudo group. Lastly, the script was deleted. This can be normal depending on if the actions are authorized. If I were an administrator the action of new users being added and given administrator privileges would be a major red flag for me and could indicate some sort of compromise.

For the log analysis on Windows it was a little bit more difficult because Event Viewer is not as transparent about what is going on. I ran the script and then created a custom view from the one minute after the script ran. That was the closest I could get.

All of the events from the first script are shown in Figure 7. It starts with the console getting ready for use. Next, a user is created named Elliot. Then the user's permissions were changed and the password was reset. Then the local group was enumerated, meaning that their groups were listed. Next, Thomas's account was created and enabled. Then the password was reset and added to a security-enabled local group. Thomas's groups were then checked. A new user named Dade was added and enabled. The password was reset and then added to the security-enabled group, then the local groups were also enumerated. Lastly, Tyler was created as a new user and enabled. The password was also changed and Tyler was added to the security-enabled group. Then Tyler's permissions were enumerated as well. Thomas and Elliot were both made administrators as well. Figure 8 shows the second script run in Event Viewer. This shows that an attempt was made to query the existence of a blank password for an account. This is just checking to see if a user has a blank password. Then, the next concerning thing that happens are all the warnings about a remote command being executed. In the details it has a Base64 encoded script, shown in Figure 9, so I decoded it and got a look at what all of the script was. The script removes the users that were added from the first script, removes their profiles and privileges, cleans up the logs, clears the PowerShell history, then exits the script. In this instance if I were an administrator, I would be alerted by the warnings that popped up for a remote command execution when the second script ran. User Account Management events could be a sign of compromise as well if this action wasn't authorized.

Between these two, I personally thought that using KSystemLog was a much better experience. It was so much easier to see what was happening and when. With

event viewer it was so unclear what was caused by the script and what was just happening in the background. I also didn't have to click on each event to see what happened. It was just all right there and visible. I think that Event Viewer could be more useful when you aren't looking for something specific and are just trying to see logs or even a wider time frame of events.

## Appendix

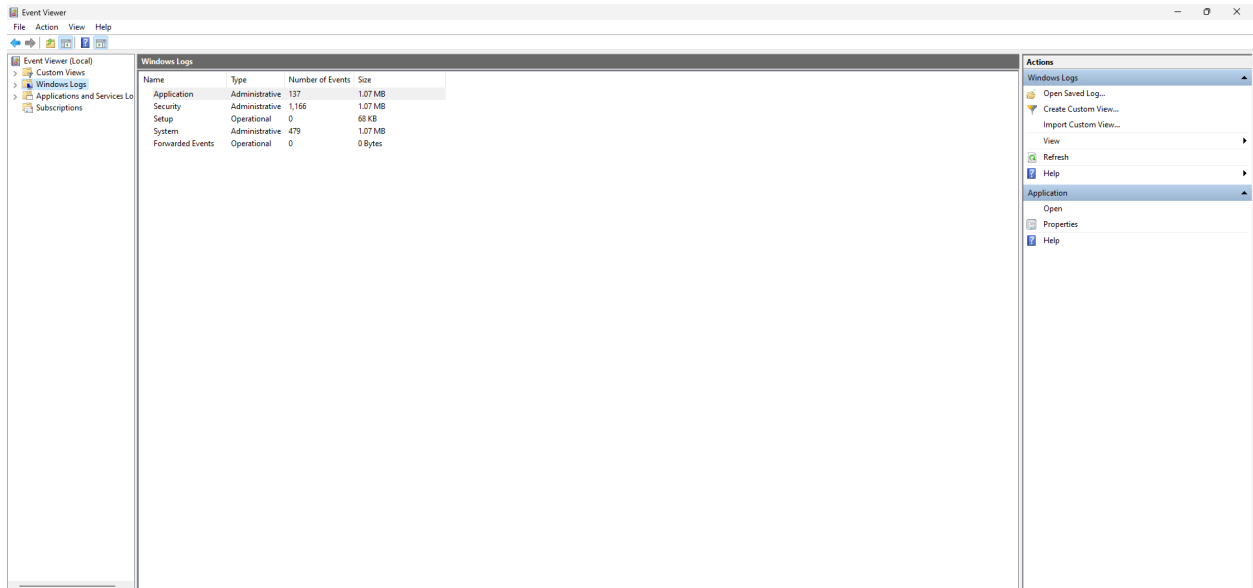


Figure 1: Windows logs

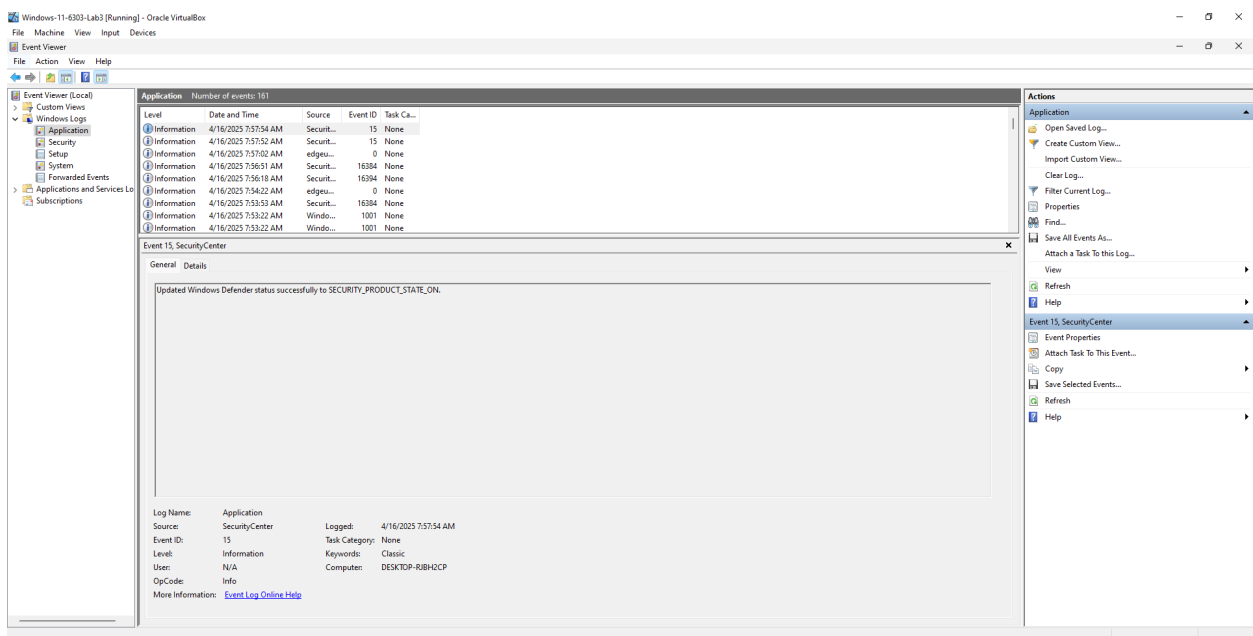


Figure 2: Event IDs

```
(kali@kali)~[/var/log]
$ journalctl
Mar 30 21:01:51 kali kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-16) 14.2.0, GNU ld >
Mar 30 21:01:51 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=aaf189f4-42db-464d-8d49-65ccc2f6cd08 ro q>
Mar 30 21:01:51 kali kernel: BIOS-provided physical RAM map:
Mar 30 21:01:51 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Mar 30 21:01:51 kali kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Mar 30 21:01:51 kali kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Mar 30 21:01:51 kali kernel: BIOS-e820: [mem 0x0000000000100000-0x0000000000dfffff] usable
Mar 30 21:01:51 kali kernel: BIOS-e820: [mem 0x0000000000dfff0000-0x0000000000dfffff] ACPI data
Mar 30 21:01:51 kali kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
Mar 30 21:01:51 kali kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Mar 30 21:01:51 kali kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Mar 30 21:01:51 kali kernel: BIOS-e820: [mem 0x0000000100000000-0x000000021fffffffff] usable
Mar 30 21:01:51 kali kernel: NX (Execute Disable) protection: active
Mar 30 21:01:51 kali kernel: APIC: Static calls initialized
Mar 30 21:01:51 kali kernel: SMBIOS 2.5 present.
Mar 30 21:01:51 kali kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Mar 30 21:01:51 kali kernel: DMI: Memory slots populated: 0/0
Mar 30 21:01:51 kali kernel: Hypervisor detected: KVM
Mar 30 21:01:51 kali kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Mar 30 21:01:51 kali kernel: kvm-clock: using sched offset of 1637985289049 cycles
Mar 30 21:01:51 kali kernel: clocksource: kvm-clock: mask: 0xffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881590591483 ns
Mar 30 21:01:51 kali kernel: tsc: Detected 2808.008 MHz processor
Mar 30 21:01:51 kali kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Mar 30 21:01:51 kali kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Mar 30 21:01:51 kali kernel: last_pfn = 0x220000 max_arch_pfn = 0x400000000
Mar 30 21:01:51 kali kernel: MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from 8 variable MTRRs
Mar 30 21:01:51 kali kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Mar 30 21:01:51 kali kernel: CPU MTRRs all blank - virtualized system.
Mar 30 21:01:51 kali kernel: last_pfn = 0x0000 max_arch_pfn = 0x400000000
Mar 30 21:01:51 kali kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
Mar 30 21:01:51 kali kernel: RAMDISK: [mem 0x28755000-0x303a1fff]
Mar 30 21:01:51 kali kernel: ACPI: Early table checksum verification disabled
lines 1-32 ... skipping ...
```

Figure 3: journalctl

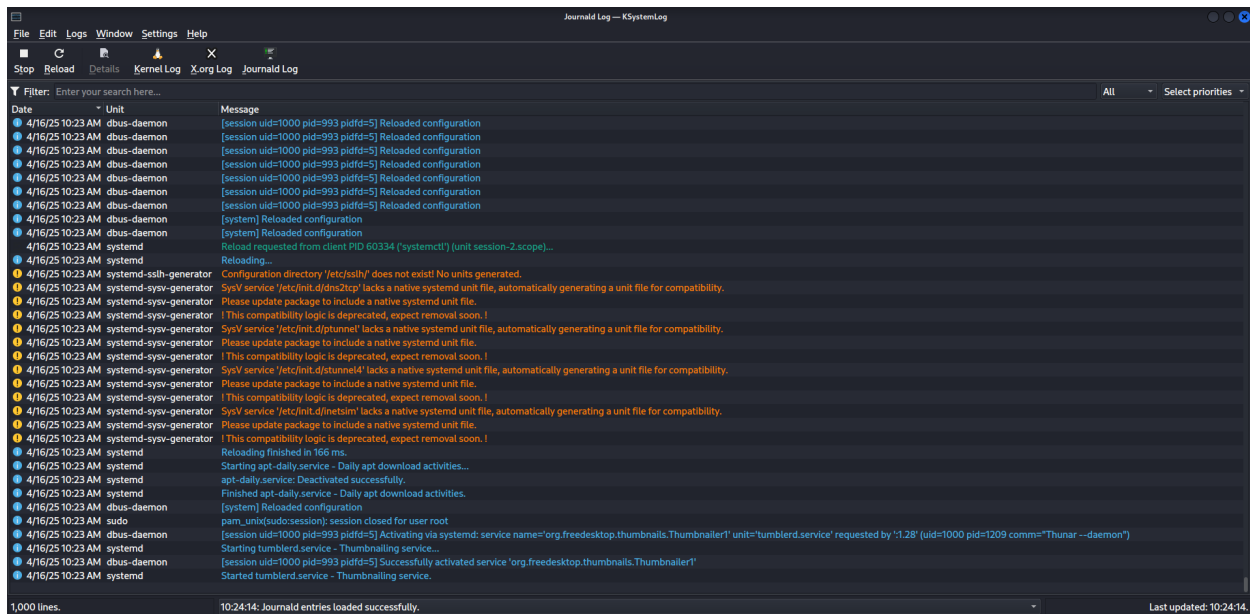


Figure 4: KSystemLog

```

4/17/25 9:35 AM sudo      kali : TTY=pts/0 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=./lab3s3.sh
4/17/25 9:35 AM sudo      pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
4/17/25 9:35 AM useradd    new group: name=Tyler, GID=1001
4/17/25 9:35 AM useradd    new user: name=Tyler, UID=1001, GID=1001, home=/home/Tyler, shell=/bin/sh, from=/dev/pts/1
4/17/25 9:35 AM useradd    new group: name=Dade, GID=1002
4/17/25 9:35 AM useradd    new user: name=Dade, UID=1002, GID=1002, home=/home/Dade, shell=/bin/sh, from=/dev/pts/1
4/17/25 9:35 AM useradd    new group: name=Elliot, GID=1003
4/17/25 9:35 AM useradd    new user: name=Elliot, UID=1003, GID=1003, home=/home/Elliot, shell=/bin/sh, from=/dev/pts/1
4/17/25 9:35 AM useradd    new group: name=Thomas, GID=1004
4/17/25 9:35 AM useradd    new user: name=Thomas, UID=1004, GID=1004, home=/home/Thomas, shell=/bin/sh, from=/dev/pts/1
4/17/25 9:35 AM chpasswd   pam_unix(chpasswd:chauthtok): password changed for Tyler
4/17/25 9:35 AM chpasswd   gkr-pam: couldn't update the login keyring password: no old password was entered
4/17/25 9:35 AM chpasswd   pam_unix(chpasswd:chauthtok): password changed for Dade
4/17/25 9:35 AM chpasswd   gkr-pam: couldn't update the login keyring password: no old password was entered
4/17/25 9:35 AM chpasswd   pam_unix(chpasswd:chauthtok): password changed for Elliot
4/17/25 9:35 AM chpasswd   gkr-pam: couldn't update the login keyring password: no old password was entered
4/17/25 9:35 AM chpasswd   pam_unix(chpasswd:chauthtok): password changed for Thomas
4/17/25 9:35 AM chpasswd   gkr-pam: couldn't update the login keyring password: no old password was entered
4/17/25 9:35 AM usermod    add 'Elliot' to group 'sudo'
4/17/25 9:35 AM usermod    add 'Elliot' to shadow group 'sudo'
4/17/25 9:35 AM usermod    add 'Thomas' to group 'sudo'
4/17/25 9:35 AM usermod    add 'Thomas' to shadow group 'sudo'
4/17/25 9:35 AM sudo      pam_unix(sudo:session): session closed for user root

```

Figure 5: First script for Kali

```

4/17/25 9:40 ... sudo      kali : TTY=pts/0 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=./lab3s4.sh
4/17/25 9:40 ... sudo      pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
4/17/25 9:40 ... sudo      root : TTY=pts/1 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/bin/bash
4/17/25 9:40 ... sudo      pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=0)
4/17/25 9:40 ... userdel    delete user 'Tyler'
4/17/25 9:40 ... userdel    removed group 'Tyler' owned by 'Tyler'
4/17/25 9:40 ... userdel    removed shadow group 'Tyler' owned by 'Tyler'
4/17/25 9:40 ... userdel    delete user 'Dade'
4/17/25 9:40 ... userdel    removed group 'Dade' owned by 'Dade'
4/17/25 9:40 ... userdel    removed shadow group 'Dade' owned by 'Dade'
4/17/25 9:40 ... userdel    delete user 'Elliot'
4/17/25 9:40 ... userdel    delete 'Elliot' from group 'sudo'
4/17/25 9:40 ... userdel    removed group 'Elliot' owned by 'Elliot'
4/17/25 9:40 ... userdel    removed shadow group 'Elliot' owned by 'Elliot'
4/17/25 9:40 ... userdel    delete 'Elliot' from shadow group 'sudo'
4/17/25 9:40 ... userdel    delete user 'Thomas'
4/17/25 9:40 ... userdel    delete 'Thomas' from group 'sudo'
4/17/25 9:40 ... userdel    removed group 'Thomas' owned by 'Thomas'
4/17/25 9:40 ... userdel    removed shadow group 'Thomas' owned by 'Thomas'
4/17/25 9:40 ... userdel    delete 'Thomas' from shadow group 'sudo'
4/17/25 9:40 ... sudo      pam_unix(sudo:session): session closed for user root
4/17/25 9:40 ... sudo      root : TTY=pts/1 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/bin/rm -f ./lab3s4.sh
4/17/25 9:40 ... sudo      pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=0)
4/17/25 9:40 ... sudo      pam_unix(sudo:session): session closed for user root
4/17/25 9:40 ... sudo      pam_unix(sudo:session): session closed for user root

```

Figure 6: Script two for Kali



[illegible]

Figure 8: Script 2 in Windows

[illegible]

Figure 9: Base64



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